PHYSICS

1)

If $y=e^x.\cot x$ then $\frac{dy}{dx}$ will be

(1) $e^x \cot x - \operatorname{cosec}^2 x$
(2) $e^x \operatorname{cosec}^2 x$
(3) $e^x[\cot x - \operatorname{cosec}^2 x]$
(4) $e^x \cot x$

2) Area of ellipse $\frac{x^2}{9}+\frac{y^2}{4}=1$ is :-

(1) 6π
(2) 12π
(3) 8π
(4) 18π

3) The radius of a circular plate increases at the rate of 0.1 cm per second. At what rate does the area increase when the radius of plate is $\frac{5}{\pi}$ cm ?

(1) 1 cm^2/s
(2) 0.1 cm^2/s
(3) 0.5 cm^2/s
(4) 2 cm^2/s

4) $\int_0^{\pi/3} \sin x\,dx$

(1) $\frac{1}{2}$
(2) 1
(3) $\frac{\sqrt{3}}{2}$
(4) 0

5) The slope of straight line $3y-6-\sqrt{3}x=0$ is

(1) $\frac{2}{\sqrt{3}}$

(2) $\sqrt{3}$

(3) $\frac{1}{\sqrt{3}}$

(4) $2\sqrt{3}$

6) cos0º.cos1º.cos2º ..... cos180º =

(1) 1

(2) –1

(3) 0

(4) $-\frac{1}{2}$

7) If $y = 2x^3 - 12x^2 + 36$ then for minimum value of y :-

(1) x = 0

(2) x = 4

(3) x = 1

(4) None

8) Value of tan(1837°) is :-

(1) 4/3

(2) 4/5

(3) 3/4

(4) 5/4

9) Find $\sqrt{101}$ (approx) :-

(1) 10.5

(2) 10.05

(3) 10.1

(4) None

10) Value of tan 74° is

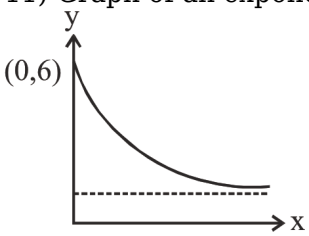
(1) $\frac{7}{25}$

(2) $\frac{13}{25}$

(3) $\frac{24}{25}$

(4) $\frac{24}{7}$

11) Graph of an exponential function $y = 2 + ae^{-x}$ is shown in figure. What is the value of a ?



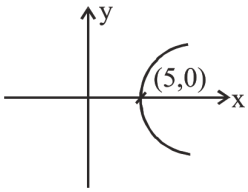
(1) 6

(2) 4

(3) 5

(4) 2

12) The equation of given curve can be :-



(1) $y = -3x^2 - 5$

(2) $y = -3x^2 + 5$

(3) $x = -3y^2 + 5$

(4) $x = 3y^2 + 5$

13) $\int_{-1}^{1} e^x \, dx$ = ?

(1) $e + \frac{1}{e}$

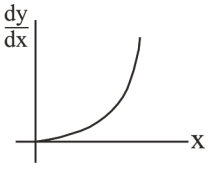
(2) $e^2 + \frac{1}{e^2}$

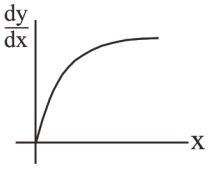
(3) $e - \frac{1}{e}$

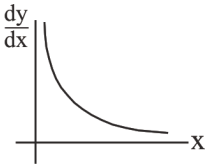
(4) $e^3 + \frac{1}{e^3}$

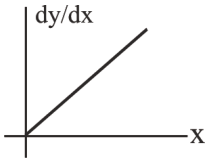
14) $y = 2x^3 + 5$ then graph between $\frac{dy}{dx}$ and x is :

(1)



(2) 

(3) 

(4) 

15) Evaluate $\int \sin^2 x \cos x \, dx.$

(1) $\frac{1}{3}\sin^3 x + C$

(2) $-\frac{1}{3}\cos^3 x + C$

(3) $\frac{1}{2}\sin^2 x + C$

(4) $-\frac{1}{2}\cos^2 x + C$

16)

Column-I		Column-II	
(A)	ε_0	(1)	$MT^{-2}A^{-1}$
(B)	μ_0	(2)	$MLT^{-3}A^{-1}$
(C)	E	(3)	$M^{-1}L^{-3}T^4A^2$
(D)	B	(4)	$MLT^{-2}A^{-2}$

	A	B	C	D
(1)	1	2	3	4
(2)	3	4	2	1
(3)	3	4	1	2
(4)	4	3	1	2

(1) 1

(2) 2

(3) 3

(4) 4

17) In a particular system the units of length, mass and time are chosen to be 10 cm, 10 g and 0.1 s respectively. The unit of force in this system will be equal to

(1) 0.1 N

(2) 1 N

(3) 10 N

(4) 100 N

18) In a system of units, the unit of mass, length and time are 1 quintal, 1 km and 1 h, respectively. In this system 1 N force will be equal to:

(1) 1 new unit

(2) 129.6 new unit

(3) 125.7 new unit

(4) 103 new unit

19)

The period T of a soap bubble under SHM is given by $T = P^aD^bS^c$, where P is pressure, D is density and S is surface tension. Then the values of a, b and c are

(1) $-\frac{3}{2}, \frac{1}{2}, 1$

(2) –1, –2, 3

(3) $\frac{1}{2}, -\frac{3}{2}$, –1/2

(4) $1, 2, \frac{3}{2}$

20) In the relation $p = \frac{\alpha}{\beta} e^{\frac{-\alpha Z}{k\theta}}$ P is pressure, Z is the distance, k is Boltzmann constant and θ is temp. The dimensional formula of β is

(1) $[M^0L^2T^0]$

(2) $[M^1L^2T^{-1}]$

(3) $[M^1L^0T^{-1}]$

(4) $[M^0L^2T^{-1}]$

21) You measure two quantities as
A = 1.0 m ± 0.2 m, B = 2.0 m ± 0.2 m. We should report correct value for $\sqrt{AB}$ as:

(1) 1.4 m ± 0.4 m

(2) 1.41m ± 0.15 m

(3) 1.4m ± 0.3 m

(4) 1.4m ± 0.2 m

22) In an experiment four quantities a, b, c and d are measured with percentage error 1%, 2%, 3% and 4% respectively. Quantity P is calculated as follows $P = \frac{a^3b^2}{cd}$ then % error in measurement of P is :-

(1) 4%

(2) 14%

(3) 10%

(4) 7%

23) A science student takes 100 observations in an experiment. Second time he takes 600 observations in the same experiment. By doing so the random error becomes :-

(1) $\frac{1}{6}$ times

(2) 6 times

(3) unchanged

(4) 36 times

24) While measuring the acceleration due to gravity by a simple pendulum, a student makes an error of 1% in the length of the pendulum and an error of 3% in the value of time period. His maximum percentage error in the measurement of g by the relation $g = 4\pi^2 (\ell/T^2)$ will be :-

(1) 2%

(2) 4%

(3) 7%

(4) 10%

25) Two resistors of resistances $R_1 = (100 \pm 3)\ \Omega$ and $R_2 = (200 \pm 4)\ \Omega$ are connected in parallel. The equivalent resistance of the parallel combination is approximately is :

(1) $(66.7 \pm 1.8)\ \Omega$

(2) $(66.7 \pm 4.0)\ \Omega$

(3) $(66.7 \pm 3.0)\ \Omega$

(4) $(66.7 \pm 7.0)\ \Omega$

26) If $a = 8 \pm 0.08$ and $b = 6 \pm 0.06$ let $x = a + b$,
$y = a - b$, $z = ab$. The correct order of % error in x, y and z is

(1) $x = y < z$

(2) $x = y > z$

(3) $x > z < y$

(4) $x < z < y$

27) $\int \frac{dx}{\sqrt{2ax - x^2}} = a^n \sin^{-1}\left[\frac{x}{a} - 1\right]$.
The value of n is :- {Use dimensional method}

(1) 0

(2) –1

(3) 1

(4) None of these

28) Zero error of an instrument introduces :

(1) Random error

(2) Systematic error

(3) Both (1) and (2)

(4) None of these

29) The order of magnitude of which quantity is Largest :-

$A = 640$

$B = 8.2 \times 10^{4}$

$C = 120 \times 10^{2}$

$D = 72000 \times 10^{-1}$

(1) A

(2) B

(3) C

(4) D

30) **Assertion (A):** - In multiplication or division, the final result should retain as many significant figures as are there in the original number with least significant figures.
Reason (R): - In addition or subtraction, the final result should retain as many decimal places as are there in the numbers with the least decimal places.

(1) Both (A) and (R) are false

(2) Both (A) and (R) are true

(3) (A) is true but (R) is false

(4) (A) is false but (R) is true

31) Length and diameter of a solid cylinder is measured using vernier callipers and screw guage having least counts 0.01 cm and 0.001 cm respectively. If measured length and radius are 10.00cm and 1.000 cm respectively, then percentage error in measurement of its volume is

(1) 0.2%

(2) 0.3%

(3) 0.4%

(4) 3%

32) A student measured the length of a rod and wrote it as 3.50 cm. Which instrument did he use to measure it ?

(1) A meter scale

(2) A vernier calliper where 10 divisions in vernier scale matches with 9 divisions in main scale and the main scale has 10 divisions in 1 cm.

(3) A screw gauge having 100 division in the circular scale and pitch as 1mm.

(4) A screw gauge having 50 divisions in the circular scale and pitch as 1mm.

33) A screw gauge with a pitch of 0.5mm and a circular scale with 50 divisions is used to measure

the thickness of a thin sheet of Aluminium. Before starting the measurement, it is found that when the two jaws of the screw gauge are brought in contact, the 45th division coincides with the main scale line and that the instrument has negative zero error. What is the thickness of the Sheet if the main scale reading is 0.5mm and the 25th division coincides with the main scale line :-

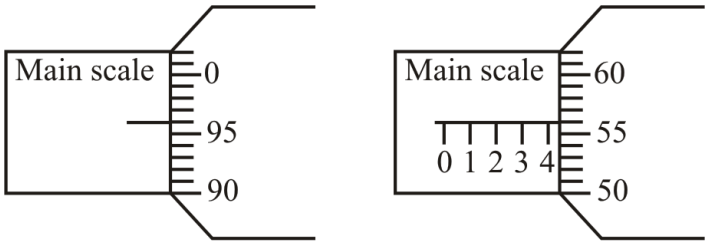
(1) 0.80 mm

(2) 0.70 mm

(3) 0.50 mm

(4) 0.75 mm

34) Two full turns of Screw gauge cover a distance of 1 mm on main scale. The total number of circular scale division are 100. The measured value of thickness of sheet will be :-



Screw Gauge

(1) 4.555 mm

(2) 4.295 mm

(3) 2.300 mm

(4) 2.245 mm

35) In an experiment, the angles are required to be measured using an instrument. 29 divisions of the main scale exactly coincide with the 30 divisions of the vernier scale. If the smallest division of the main scale is half-a-degree(= 0.5°), then the least count of the instrument is

(1) One degree

(2) Half degree

(3) One minute

(4) Half minute

36) The diameter of a cylinder is measured using a Vernier callipers with no zero error. It is found that the zero of the Vernier scale lies between 5.10 cm and 5.15 cm of the main scale. The Vernier scale has 50 divisions equivalent to 2.45 cm. The 24th division of the Vernier scale exactly coincides with one of the main scale divisions. The diameter of the cylinder is :-

(1) 5.112 cm

(2) 5.124 cm

(3) 5.136 cm

(4) 5.148 cm

37) Find $F_{net} = GMm\left[\frac{1}{r^2} + \frac{1}{2r^2} + \frac{1}{4r^2} + \ldots\ldots\ldots\ldots \text{upto } \infty\right]$

(1) $\frac{2GMm}{r^2}$

(2) $\frac{3GMm}{r^2}$

(3) $\frac{4GMm}{r^2}$

(4) $\frac{5GMm}{r^2}$

38) A vector of length ℓ is turned through the angle θ about its tail. What is the change in the position vector of its head :-

(1) $\ell \cos(\theta/2)$

(2) $2\ell \sin(\theta/2)$

(3) $2\ell \cos(\theta/2)$

(4) $\ell \sin(\theta/2)$

39) Find unit vector perpendicular to $\vec{A}$ and $\vec{B}$ where $\vec{A} = \hat{i} - 2\hat{j} + \hat{k}$, $\vec{B} = \hat{i} + 2\hat{j}$

(1) $\frac{2\hat{i} + \hat{j} + 4\hat{k}}{\sqrt{21}}$

(2) $\frac{-2\hat{i} + \hat{j} + 4\hat{k}}{\sqrt{21}}$

(3) $\frac{-2\hat{i} - \hat{j} + 4\hat{k}}{\sqrt{21}}$

(4) $\frac{2\hat{i} + \hat{j} + \hat{k}}{\sqrt{5}}$

40) The maximum and minimum resultant of two forces acting at a point are 10N and 6N respectively. If each force is increased by 3N, find the resultant of new forces when acting at a point at an angle of 90° with each-other :-

(1) $\sqrt{146}\,N$

(2) 11 N

(3) $\sqrt{70}\,N$

(4) 8 N

41) The unit vector parallel to the resultant of the vectors
$\vec{A} = 2\hat{i} - 4\hat{j} - 6\hat{k}$ & $\vec{B} = 4\hat{i} + 3\hat{j} + 3\hat{k}$

(1) $\frac{1}{\sqrt{46}}(6\hat{i} + \hat{j} + 3\hat{k})$

(2) $\frac{1}{\sqrt{46}}(6\hat{i} - \hat{j} - 3\hat{k})$

(3) $\frac{1}{\sqrt{46}}(2\hat{i} - 4\hat{j} - 6\hat{k})$

(4) $\frac{1}{\sqrt{46}}(4\hat{i}+3\hat{j}+3\hat{k})$

42) Given $|\vec{A}_1| = 2,\ |\vec{A}_2| = 3$ and $|\vec{A}_1 + \vec{A}_2| = 3$. Find the value of $(\vec{A}_1 + 2\vec{A}_2) \cdot (3\vec{A}_1 - 4\vec{A}_2)$

(1) –64
(2) 60
(3) –60
(4) 64

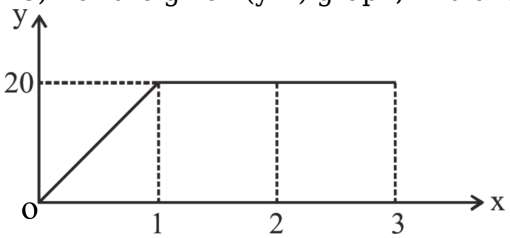
43) The vector component of vector $\vec{A} = 3\hat{i}+4\hat{j}+5\hat{k}$ along vector $\vec{B} = \hat{i}+\hat{j}+\hat{k}$ is

(1) $2\hat{i}+2\hat{j}+2\hat{k}$
(2) $3\hat{i}+3\hat{j}+3\hat{k}$
(3) $4\hat{i}+4\hat{j}+4\hat{k}$
(4) $5\hat{i}+5\hat{j}+5\hat{k}$

44) If $\hat{i}, \hat{j}$ and $\hat{k}$ represent unit vectors along the x, y and z-axes respectively, then the angle θ between the vectors $\hat{i}+\hat{j}+\hat{k}$ and $\hat{i}+\hat{j}$ is equal to :-

(1) $\sin^{-1}\left(\frac{1}{\sqrt{3}}\right)$
(2) $\sin^{-1}\left(\sqrt{\frac{2}{3}}\right)$
(3) $\cos^{-1}\left(\frac{1}{\sqrt{3}}\right)$
(4) 90°

45) For the given (y–x) graph, find average value of y over an interval $0 \le x \le 3$:-



(1) 10
(2) $\frac{50}{3}$
(3) $\frac{20}{3}$
(4) 20

CHEMISTRY

1) Number of neutrons in 4 g H_2 is nearly :-

(1) 1.2×10^{24}
(2) 6×10^{23}
(3) 3×10^{23}
(4) None

2) The quantum number which is not obtained from the solution of Schrodinger equation is

(1) Principle quantum number
(2) Azimuthal quantum number
(3) Spin quantum number
(4) Magnetic quantum number

3) The total number of atomic orbitals in the third energy level of an atom is

(1) 9
(2) 8
(3) 18
(4) 32

4) One litre of a gas at STP has 1.25 g mass. It can possibly be :-

(1) C_2H_2
(2) CO
(3) O_2
(4) CH_3

5) A mixture of gases contains CH_4 and SO_3 gases in ratio of 2 : 5 (w/w). What is the molar ratio of the two gases in the mixture :-

(1) 1 : 2
(2) 2 : 1
(3) 1 : 4
(4) 4 : 1

6) If 10^{21} molecules are removed from 200 mg of CO_2, then the number of moles of CO_2 left is :-

(1) 2.88×10^{-3}
(2) 1.66×10^{-3}
(3) 4.54×10^{-3}
(4) 1.66×10^{-2}

7) The density of CH_4 at NTP is :-

(1) 0.71 g L^{-1}

(2) 0.23 g L^{-1}

(3) 0.15 g L^{-1}

(4) 0.52 g L^{-1}

8) The vapour density of a gaseous hydrocarbon is 36. If hydrocarbon is C_nH_{2n+2} then n is :-

(1) 2

(2) 4

(3) 5

(4) 6

9) Four 1-L flasks are separately filled with the gas hydrogen, helium, oxygen and ozone at the same room temperature and pressure. The ratio of total number of atoms of these gases present in the different flasks would be?

(1) 1 : 1 : 1 : 1

(2) 1 : 2 : 2 : 3

(3) 2 : 1 : 2 : 3

(4) 1 : 2 : 2 : 1

10) **Statement I :** In ground state configuration of $_{24}Cr$ number of electrons with $n + l \geq 3$ is 20.
Statement II : In $_{36}Kr$ total number of electrons with m = 0 and s = +1/2 is 8.

(1) Both statements I and II are incorrect

(2) Both statements I and II are correct

(3) Only statement I is correct

(4) Only statement II is correct

11) How much Cu can be obtained from
100 g of $CuSO_4$. (Atomic weight Cu = 63.5)

(1) 20 g

(2) 40 g

(3) 60 g

(4) 80 g

12) Amount of oxygen required for combustion of
1 Kg of a mixture of butane and isobutane is :–

(1) 1.8 Kg

(2) 2.7 Kg

(3) 4.5 Kg

(4) 3.58 Kg

13) 0.2 litre of gaseous hydrocarbon on combustion produces 800mL of $CO_2(g)$ and 1000mL of $H_2O(g)$ under same conditions of temperature and pressure. Calculate molecular formula of

hydrocarbon

(1) C_3H_8

(2) C_4H_{10}

(3) C_5H_{12}

(4) None

14) If a hydrocarbon having 80% carbon and vapour density of compound is 30. What is the molecular formula is the compound ?

(1) CH_3

(2) C_2H_6

(3) C_4H_{12}

(4) CH_3COOH

15) How many moles of $KClO_3$ to be heated to produce 11.2 litre of O_2 at NTP.
$2KClO_3 \xrightarrow{\Delta} 2KCl + 3O_2(g)$

(1) 0.66

(2) 0.33

(3) 1.33

(4) 0.13

16) To completly neutralise 12 gm NaOH, 9.8 g of Tribasic acid is required, then molar weight of acid is :

(1) 49 gm

(2) 98 gm

(3) 60 gm

(4) 72 gm

17) If 0.5 mol of $CaBr_2$ is mixed with 0.2 mol of K_3PO_4 then the maximum number of moles of $Ca_3(PO_4)_2$ obtained will be :- $3CaBr_2 + 2K_3PO_4 \rightarrow Ca_3(PO_4)_2 + 6KBr$

(1) 0.5

(2) 0.2

(3) 0.7

(4) 0.1

18) If 8 mL of uncombined O_2 remain after exploding O_2 with 4 mL of hydrogen, the volume in mL of O_2 originally was :

(1) 12

(2) 2

(3) 10

(4) 4

19) At NTP, In a container 100 mL of N_2 and 100 mL of H_2 are mixed together. Find out the produced volume of NH_3.

(1) 40 mL
(2) 52.8 mL
(3) 60 mL
(4) 66.6 mL

20) An aqueous solution of 6.3 g of oxalic acid dihydrate is made upto 250 mL. The volume of 0.1 N NaOH required to completely neutralise 10 mL of oxalic acid is:

(1) 40 mL
(2) 20 mL
(3) 10 mL
(4) 4 mL

21) Which one of the following pairs of compounds illustrates the law of multiple proportions?

(1) H_2O, Na_2O
(2) MgO, Na_2O
(3) Na_2O, BaO
(4) $SnCl_2$, $SnCl_4$

22) Two oxides of a metal contain 22.22% and 30% oxygen by mass respectively. If the formula of the first oxide is MO, then the formula of the second oxide is :-

(1) MO
(2) MO_2
(3) M_2O_3
(4) M_2O

23) Which is not true with respect to cathode rays ?

(1) A stream of electrons
(2) Charged particles
(3) Can be deflected by mgnetic field
(4) Move with same speed as that of light

24) A 100 watt bulb emits light of λ = 400 Å, the number of photons/sec emitted by the bulb will be-

(1) 5×10^{16}
(2) 2×10^{19}
(3) 20×10^{29}
(4) 5×10^{15}

25) **Statement I :** According to Heisenberg's product of uncertainty in measurement of velocity and uncertainty in measurement of position is $\geq h/4\pi$.
Statement II : $Fe_2O_3 + Al \longrightarrow Fe + Al_2O_3 + Q$ ratio of masses of Fe_2O_3 and Al taken to given maximum energy is 80 : 27.

(1) Both statements I and II are correct
(2) Both statements I and II are incorrect
(3) Only statement I is correct
(4) Only statement II is correct

26) For an electron, with n = 3 has only one radial node. The orbital angular momentum of the electron will be

(1) 0
(2) $\sqrt{6}\frac{h}{2\pi}$
(3) $\sqrt{2}\frac{h}{2\pi}$
(4) $3\left(\frac{h}{2\pi}\right)$

27) **Statement I :** On diluting solution, it's normality and molarity changes but molality remains constant.
Statement II : The total number of atoms in 4 amu He and 36 amu H_2O will be $7N_A$.

(1) Only statement I is correct
(2) Only statement II is correct
(3) Both statements I and II are incorrect
(4) Both statements I and II are correct

28) Two radiations have the ratio of energies 3 : 8. If first radiation has the wave number 150 cm^{-1} then wave number of second radiation will be :-

(1) 200 cm^{-1}
(2) 20 cm^{-1}
(3) 56.25 cm^{-1}
(4) 400 cm^{-1}

29) **Assertion :** $2p_x$, $2p_y$ and $2p_z$ each have one nodal plane.
Reason : These orbital are degenrate orbitals.

(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.
(2) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.
(3) Assertion is true but Reason is false.
(4) Both Assertion and Reason are false.

30) In which of the following radius of first orbit is maximum ?

(1) H

(2) He^{+}

(3) Li^{+2}

(4) Be^{+3}

31) A hydrogen atom is in an excited state **'n'** is ionised by the absorption of radiation with a photon energy of 1 eV. What is the minimum value of **'n'** for excited state of the atom :-

(1) 1

(2) 2

(3) 3

(4) 4

32) Identify the incorrect statement from the following -

(1) Value of Azimuthal Quantum number is same for 3d, 4d and 5d subshells

(2) Electron distribution in d_{xy} orbital is in between the axis and in Px orbital is along the axis.

(3) ψ_{520} represents $5dz^2$ orbitals

(4) In hydrogen atom, energy of 5d and 4f subshells are same

33) In which of the following transition maximum energy radiation will be emitted for He^{+} ion

(1) n = 5 to n = 2

(2) n = 3 to n = 1

(3) n = 7 to n = 3

(4) n = 8 to n = 4

34) Hydrogen atoms are excited to n = 4 state. In the spectrum of emitted radiation, number of lines in the ultraviolet and visible regions are respectively :-

(1) 3 and 1

(2) 1 and 3

(3) 2 and 3

(4) 3 and 2

35) The wavelength of third line of the Balmer series for H atom is:-

(1) $\frac{21}{100R}$

(2) $\frac{21R}{100}$

(3) $\frac{100R}{21}$

(4) $\frac{100}{21R}$

36) The shortest wavelength of He^{+} in Balmer series is x, then longest wavelength in the Paschen

series of Li^{+2} is :-

(1) $\frac{36x}{5}$

(2) $\frac{16x}{7}$

(3) $\frac{9x}{5}$

(4) $\frac{5x}{9}$

37) The de Broglie wavelength of a 66 kg man sking down Kufri Hill in Shimla at 1×10^{3} m sec^{-1} is

(1) 1×10^{-36} m

(2) 1×10^{-37} m

(3) 1×10^{-38} m

(4) 1×10^{-39} m

38) If the de-broglie wave length of an electron revolving in 2nd orbit of H-atom is x. Then radius of the orbit is :-

(1) $\frac{x}{\pi}$

(2) $\frac{2x}{\pi}$

(3) $\frac{x}{2\pi}$

(4) $\frac{4x}{\pi}$

39) The uncertainty in the position of a moving bullet of mass 10 gm is 10^{-5} m. Calculate the uncertainty in its velocity :-

(1) 5.2×10^{-28} m/sec

(2) 3×10^{-28} m/sec

(3) 5.2×10^{-22} m/sec

(4) 3×10^{-22} m/sec

40) Which have the same number of s-electrons as the d-electrons in Fe^{2+} ?

(1) Li

(2) Na

(3) N

(4) P

41) The probability of finding an electron residing in a p_x orbital is zero

(1) in the yz plane

(2) in the xy plane

(3) in the xz plane

(4) in the x-direction

42) The number of electrons of same spin with
n + ℓ = 5 are :-

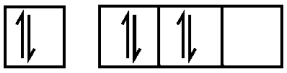
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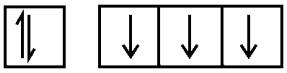
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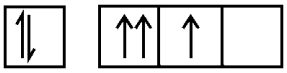
(3) 12

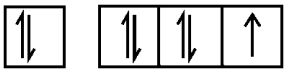
(4) 8

43) The orbital diagram in which both the Pauli's exclusion principle and Hund's rule are violated is:-

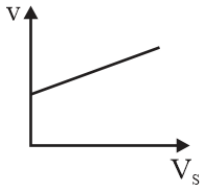
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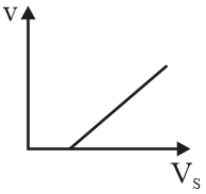
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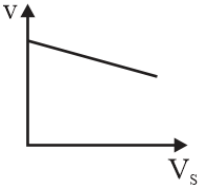
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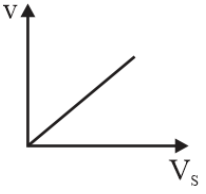
(4) 

44) Graph of incident frequency (v) with stopping potential (v_s) in photoelectric effect is:

(1) 

(2) 

(3) 

(4) 

45)

Number of electrons in Cr which have l + m = 0

(1) 2

(2) 4

(3) 13

(4) 14

BIOLOGY

1) **Statement-I :** Mitosis is the equational division in which the chromosome number in the parent and progeny cell remains the same.
Statement-II : In sexual reproduction when the two gametes fuse the chromosome number is restored to the value in the parent.

(1) Both statements are false.

(2) Both statements are true.

(3) Statement-I is correct but statement-II is incorrect.

(4) Statement-I is incorrect but statement-II is correct.

2) **Assertion (A):** Meiosis occurs in somatic cells of multicellular organisms.
Reason (R): Meiosis is responsible for growth and repair of body tissues.

(1) Both Assertion and Reason are true, and Reason is the correct explanation of Assertion.

(2) Both Assertion and Reason are true, but Reason is not the correct explanation of Assertion.

(3) Assertion is true, but Reason is false.

(4) Both Assertion and Reason are false.

3) **Assertion:** Crossing over is an enzyme mediated process.
Reason: In crossing over recombinase enzyme helps in exchange of genetic material.

(1) Both Assertion and Reason are true, and Reason is the correct explanation of Assertion.

(2) Both Assertion and Reason are false, but Reason is not the correct explanation of Assertion.

(3) Assertion is true, but Reason is false.

(4) Both Assertion and Reason are false.

4) **Statement-I:** Chromosomes become gradually visible under light microscope during leptotene stage.
Statement-II: In oocytes of some vertebrates, diplotene can last for months or years.

(1) Both statement-I and II are false.

(2) Both statement-I and II are true.

(3) Statement-I is true but statement-II is false.

(4) Statement-II is true but statement-I is false.

5) **Statement I :** The complete disintegration of the nuclear envelope marks the start of anaphase.
Statement II : Meiosis involves two sequential cycles of nuclear and cell division called meiosis I and meiosis II but only a single cycle of DNA replication.

(1) Statement I and statement II both are true

(2) Statement I is true but statement II is false

(3) Statement I is false but statement II is true

(4) Statement I and statement II both are false

6) **Statement-I :** During G_0-phase of cell cycle, the cell is metabolically inactive.
Statement-II : Heart cells enter G_0-phase and divide on when it is required.

(1) Both Statement-I and Statement-II are incorrect.

(2) Statement-I is correct but Statement-II is incorrect.

(3) Statement-I is incorrect but Statement-II is correct.

(4) Both Statement-I and Statement-II are correct.

7) **Assertion :** Meiotic division leads to variations that are very important for the process of evolution.
Reason : Meiosis results in reduction of chromosome number by half.

(1) Both assertion and reason are true but reason is the correct explanation of assertion.

(2) Both assertion and reason are true but reason is not the correct explanation of assertion.

(3) Assertion is true but reason is false.

(4) Both assertion and reason are false.

8) **Assertion :** G_1 phase corresponds to the interval between mitotic phase and phase in which initiation of DNA replication takes place.
Reason : During G_1 phase the cell is metabolically inactive and continuously grows but does not replicate its DNA.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

9) **Assertion :** The individual chromosomes can no longer be seen and set of chromatin material tends to collect at each of the two poles in telophase.
Reason : The chromosomes that have reached their respective poles decondense and lose their individuality.

(1) Both assertion and reason are correct but reason is not the correct explanation of assertion.

(2) Assertion is correct but reason is not correct.

(3) Assertion is not correct but reason is correct.

(4) Both assertion and reason are correct and reason is the correct explanation of assertion.

10) **Assertion :** Homologous chromosomes separate during anaphase I of meiosis.
Reason : Homologous chromosomes pair during prophase II.

(1) Both Assertion and Reason are true, and Reason is the correct explanation of Assertion.

(2) Both Assertion and Reason are true, but Reason is not the correct explanation of Assertion.

(3) Assertion is true, but Reason is false.

(4) Assertion is false, but Reason is true.

11) **Assertion (A):** During anaphase II of meiosis II, sister chromatids separate and move toward opposite poles.
Reason (R): During anaphase II of meiosis II, the centromeres split and microtubules shorten, pulling the chromatids apart.

(1) Both A and R are true and R is the correct explanation of A

(2) Both A and R are true but R is not the correct explanation of A

(3) A is true but R is false

(4) A is false but R is true

12) **Assertion (A):** Chiasmata are X-shaped structures that are visible during diplotene, indicating the sites of crossovers at pachytene.
Reason (R): The synaptonemal complex dissolves at the beginning of diplotene, allowing homologous chromosomes to separate except at the crossover points.

(1) Both A and R are true and R is the correct explanation of A

(2) Both A and R are true but R is not the correct explanation of A

(3) A is true but R is false

(4) A is false but R is true

13) **Assertion (A):** Mitotic division in meristematic tissues such as apical and lateral cambium contributes to the continuous growth of plants.
Reason (R): Cell division is essential to maintain the nucleo-cytoplasmic ratio during cell growth.

(1) Both A and R are true and R is the correct explanation of A

(2) Both A and R are true but R is not the correct explanation of A

(3) A is true but R is false

(4) A is false but R is true

14) **Statement I :** The plane of aligment of the chromosomes at metaphase is refferred to as the metaphase plate.
Statement II : The metaphase plate is the plane where chromosomes line up during metaphase due to the attachment of spindle fibres to kinetochores.

(1) Both statements I and II are incorrect

(2) Both statements I and II are correct

(3) Only statement I is correct

(4) Only statement II is correct

15) **Assertion (A):** In animals, mitotic cell division typically occurs in diploid somatic cells, but **exceptions** like mitosis in haploid cells of male honey bees do exist.
Reason (R): Plants, unlike animals, can undergo mitotic divisions in both haploid and diploid cells.

(1) Both A and R are true and R is the correct explanation of A

(2) Both A and R are true but R is not the correct explanation of A

(3) A is true but R is false

(4) A is false but R is true

16) **Statement-A:** In sexually reproducing organism meiosis ensures the production of haploid phase in their life cycle but fertilisation restores diploid phase.
Statement-B: During pachytene stage chromosomes start pairing together and this process of association is called synapsis.

(1) Only statement A is correct

(2) Only statement B is correct

(3) Both statements A and B are correct

(4) Both statements A and B are incorrect

17) **Assertion (A) :-** If the initial amount of DNA is denoted as 2C then it is increased to 4C by the end of S phase.
Reason (R) :- DNA replication takes place in S phase and amount of DNA also gets doubled.

(1) Both A and R are correct but R is NOT the correct explanation of A.

(2) A is correct but R is not correct.

(3) A is not correct but R is correct.

(4) Both A and R are correct and R is the correct explanation of A.

18) **Statement-I :-** Meiosis II is initiated after the parental chromosomes have replicated to produce identical sister chromatids.
Statement-II :- Although in many cases the chromosomes do undergo some dispersion, but they do not reach the extremely extended state of the interphase nucleus in telophase-I.

(1) Both Statement-I and Statement-II are false.

(2) Statement-I is true but Statement-II is false.

(3) Statement-I false but Statement-II is true.

(4) Both Statement-I and Statement-II are true.

19) Match the column I with column II, and choose the **correct** combination from the options given.

Column I		**Column II**	
a.	Formation of synaptonemal complex	1.	Zygotene
b.	Recombination nodules	2.	Pachytene
c.	Activation of recombinase enzyme	3.	Diplotene
d.	Terminalisation of chiasmata	4.	Diakinesis

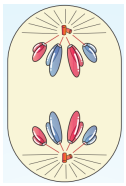
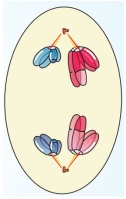
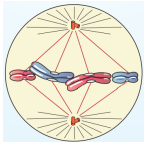
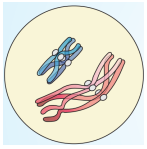
(1) a → 1; b → 2; c → 3; d → 4

(2) a → 2; b → 1; c → 3; d → 4

(3) a → 1; b → 2; c → 4; d → 3

(4) a → 1; b → 2; c → 2; d → 4

20) Match the Column-I with Column-II and select the correct option from options given below

	Column-I		Column-II
(A)		(P)	Anaphase of meiosis-I
(B)		(Q)	Metaphase of mitosis
(C)		(R)	Prophase of meiosis- I
(D)		(S)	Anaphase of mitosis

(1) A-S, B-P, C-Q, D-R

(2) A-S, B-P, C-R, D-Q

(3) A-P, B-Q, C-R, D-S

(4) A-P, B-R, C-Q, D-S

21) **Statement-I:** Prokaryotic flagella is structurally different from eukaryotic flagella.
Statement-II: The core of cilia & flagella is called axoneme.

(1) Statement I is correct, Statement II is incorrect

(2) Both Statements I and II are correct

(3) Statement I is incorrect, Statement II is correct

(4) Both Statements I and II are incorrect

22) **Statement-I :-** Nonpolar tail of saturated hydrocarbon is protected from aqueous environment in cell membrane.
Statement-II :- In cell membrane major lipids are phospholipids that are arranged in a bilayer.

(1) Both statements are correct.

(2) Both statements are not correct.

(3) Statement-I is correct, statement-II is incorrect.

(4) Statement-I is incorrect, statement-II is correct.

23) Select the **correct** option.

	Characters	Prokaryotes	Eukaryotes

(1)	Multiplication	Rapid	More rapid
(2)	Chromosomes	True	False
(3)	Compartmentalisation	Absent	Present
(4)	Ribosomes	Absent	70S

(1) 1

(2) 2

(3) 3

(4) 4

24) **Statement-I :** Plasmid DNA confers certain unique phenotypic characters to bacteria.
Statement-II : Plasmid DNA is used to monitor bacterial transformation with foreign DNA.

(1) Both Statement I and II correct.

(2) Both Statement I and II incorrect.

(3) Only Statement I correct.

(4) Only statement II correct.

25) Which of the following statement is **correct** ?
(a) Matthis Schleiden observed that all plants are composed of different kind of cells which form the tissue of the plant.
(b) Schwann, concluded that the presence of cell wall is unique character of the plant cells.
(c) Schleiden and Schwann together formulated the cell theory.
(d) Schleiden explained that cell membrane is the delimiting structure of plant cell.
Option :

(1) a, b, c and d

(2) Only b, c and d

(3) Only a, b and c

(4) Only a and d

26) Which of the following statements are **incorrect** with respect of Golgi apparatus ?
A. The cis and the trans faces of the organelle are entirely different, but inter connected.
B. The golgi cisternae are concentrically arranged near the nucleus with distinct convex cis or maturing face and concave trans or forming face.
C. It is the important site of formation of glycoprotein and glycolipids.
D. A number of proteins are modified in the cisternae of golgi complex before they are released from cis face.

(1) A and B only

(2) B and C only

(3) B and D only

(4) A and C only

27) **Statement - I :** The quasi-fluid nature of proteins enables lateral movement of lipids within the overall bilayer.
Statement - II : The polar molecules can pass through the non-polar lipid bilayer via carrier

protein.

(1) Statement - I is incorrect but statement - II is correct

(2) Both Statement - I and Statement - II are correct

(3) Both Statement - I and Statement - II are incorrect

(4) Statement - I is correct but Statement - II is incorrect

28) **True** statements for endomembrane system is:
A. Function of ribosomes are coordinated with ER so ribosomes are part of endomembrane system
B. Mitochondria and chloroplast are not the part of endomembrane system
C. Different cell organelles which have coordinated functions are included in endomembrane system
D. Vacuole is not a part of endomembrane system

(1) A, B, C and D

(2) Only B and D

(3) Only B and C

(4) Only A, B and C

29) Which of the following is **not** true about fimbriae?

(1) Surface structures of bacteria and do not play any role in motility

(2) They are known to help attach the bacteria to rocks in streams

(3) They are small bristle like fibres sprouting out of the bacterial cell

(4) They form pellicle

30) **Statement-I :** Algae have cell wall which is made of cellulose, galactans, mannans and minerals like calcium carbonate.
Statement-II : Secondary cell wall is formed on the outer side of primary cell wall toward middle lamella.

(1) Both statement-I and statement-II are correct.

(2) Both statement-I and statement-II are incorrect.

(3) Statement-I is correct and statement-II is incorrect.

(4) Statement-I is incorrect but statement-II is correct.

31) Mitochondria :-
(a) Are easily visible under the microscope (without specifically stained).
(b) Are typically sausage-shaped or cylindrical.
(c) Are double membrane bound structures.
(d) Have two aqueous compartments.

(1) (a), (d) are correct & (b), (c) are incorrect

(2) (a), (b) are correct & (c), (d) are incorrect

(3) (a) is incorrect & (b), (c), (d) are correct

(4) (a), (d) are incorrect & (b), (c) are correct

32) Which of the following statements are **correct** with respect to Golgi apparatus?

A) Discovered by Camillo Golgi
B) It produces cellular energy in the form of ATP
C) It modifies the protein synthesized by ribosomes on ER
D) Double membrane bound structure
E) The golgi apparatus principally performs the function of packaging materials, to be delivered either to the intra-cellular targets or secreted outside the cell.

(1) (B) and (C) only
(2) (A), (C) and (E) only
(3) (A), (B) and (D) only
(4) (D) and (E) only

33) Which of the following is an **incorrect** statement ?

(1) Sieve tube cells of vascular plants contain more than one nucleus
(2) Microbodies are present both in plant and animal cells
(3) The perinuclear space forms a barrier between the materials present inside the nucleus and that of the cytoplasm
(4) Nuclear pores act as passages for proteins and RNA molecules in both directions between nucleus and cytoplasm

34) Which of the following statements is **not** correct?

(1) In *Amoeba* contractile vacuole is important for osmoregulation and excretion
(2) The hydrolytic enzymes of lysosomes are active at the acidic pH
(3) Lysosomes are membrane bound vesicular structures
(4) Lysosomes are formed by the process of packaging in the endoplasmic reticulum only

35) **Assertion :-** The content of nucleolus is continuous with the rest of the nucleoplasm.
Reason :- Nucleolus is not a membrane bound structure.

(1) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.
(2) **Assertion** is true but **Reason** is false.
(3) **Assertion** is false but **Reason** is true.
(4) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

36) Find **incorrect** statement with regard to centrosome and centrioles: -
(a) Centrosomes are found in higher plants
(b) In centrosome, both centrioles lie parallel to each other in which each has an organization like the cartwheel
(c) Centrioles are made up of nine unevenly spaced peripheral fibrils of tubulin
(d) The central part of the distal region of the centriole is proteinaceous and called the Hub
(e) Proteinaceous radial spokes connect hub to peripheral triplets

(1) a, b
(2) Only b
(3) a, b, c,d

(4) All are incorrect

37) Microtubule, Microfilaments and intermediate filaments constitute-

(1) Elaborate network of filamentous proteinaceous structure present in cytoplasm.

(2) Dense granular structure present in cytoplasm which were discovered by George Palade in 1953.

(3) Network of tiny tubular double membranous structures scattered in cytoplasm

(4) All of these

38) Match the following:

Column - I		**Column - II**	
(i)	Plasmodesmata	(a)	Nuclear matrix
(ii)	Kinetochores	(b)	Thick and tough glycocalyx
(iii)	Nucleoplasm	(c)	Disc shaped structure on the surface of centromere
(iv)	Capsule	(d)	Cytoplasmic connections

(1) (i) - (d), (ii) - (a), (iii) - (c), (iv) - (b)

(2) (i) - (c), (ii) - (a), (iii) - (d), (iv) - (b)

(3) (i) - (d), (ii) - (c), (iii) - (a), (iv) - (b)

(4) (i) - (b), (ii) - (c), (iii) - (a), (iv) - (d)

39) **Statement-I :** Plastids are found in all plant cells and in Euglenoides.
Statement-II : Majority of the chloroplasts of the green plants are found in the mesophyll cells of the leaves.

(1) Statement-I and statement-II both are true

(2) Statement-I is true and statement-II is false

(3) Statement-I is false and statement-II is true

(4) Statement-I and statement-II both are false

40) Unicellular organisms
A) Are incapable of independent existence
B) Are capable of independent existence
C) Are incapable of performing the essential functions of life
D) Are capable of performing the essential functions of life

(1) All are incorrect

(2) All are correct

(3) Both B & D are correct

(4) Both A & C are correct

41) **Assertion:** Centriole form the basal body of cilla or flagella.

Reason: Nucleolus is a site for inactive RNA synthesis.

(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.
(2) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.
(3) Assertion is true but Reason is false.
(4) Both Assertion and Reason are false.

42) **Statement I:** The extensions of plasma memrbane in form of vesicles, tubules, and lamellae in cells help in processes like cell wall formation, DNA replication and distribution to daughter cells.
Statement II: In prokaryotes like cyanobacteria, chromatophores are membranous extensions that help in motility.

(1) Statement I is false but Statement II is true.
(2) Both Statement I and Statement II are true.
(3) Both Statement I and Statement II are false.
(4) Statement I is true but Statement II is false.

43) **Statement I :** The axoneme of cilia and flagella is composed of microtubules arranged in a 9+2 pattern.
Statement II : The 9+2 arrangement refers to nine pairs of peripheral microtubules and two central microtubules.

(1) Both statements I and II are correct
(2) Both statements I and II are incorrect
(3) Only statement I is correct
(4) Only statement II is correct

44) **Assertion (A) :** Chromoplasts contain fat-soluble carotenoid pigments which impart yellow, orange, or red color to plant parts.
Reason (R) : Leucoplasts are colourless plastids that store nutrients such as carbohydrates, oils, fats, and proteins.

(1) Both A and R are true and R is the correct explanation of A
(2) Both A and R are true but R is not the correct explanation of A
(3) A is true but R is false
(4) A is false but R is true

45) **Statement I :** Reductionist Biology involves understanding the molecular basis of physiological processes like digestion, excretion, and memory.
Statement II : In Reductionist Biology, the concepts and techniques of physics and chemistry are applied to study biological processes.

(1) Both statements I and II are correct
(2) Both statements I and II are incorrect
(3) Only statement I is correct
(4) Only statement II is correct

46) **Statement I :** Ligaments are dense irregular tissue.
Statement II : Cartilage is dense regular tissue.

(1) Both statements I and II are correct
(2) Both statements I and II are incorrect
(3) Only statement I is correct
(4) Only statement II is correct

47) Match List I with List II.

	List - I		**List - II**
(A)	Mast cells	(i)	Ciliated epithelium
(B)	Inner surface of bronchiole	(ii)	Areolar connective tissue
(C)	Blood	(iii)	Cuboidal epithelium
(D)	Tubular parts of nephron	(iv)	Specialised connective tissue

(1) A-i, B-ii, C-iv, D-iii
(2) A-ii, B-iii, C-i, D-iv
(3) A-ii, B-i, C-iv, D-iii
(4) A-iii, B-iv, C-ii, D-i

48) Which of the following is not a connective tissue?

(1) Bone
(2) Neuron
(3) Lymph
(4) Areolar tissue

49) Match List-I with List-II:

	List - I		**List - II**
(A)	Gall Bladder	(i)	Dense regular connective tissue
(B)	Goblet cell	(ii)	Looose connectie tissue
(C)	Ligament	(iii)	Glandular tissue
(D)	Areolar tissue	(iv)	Columnar epithelium

(1) A-ii, B-i, C-iv, D-iii
(2) A-iii, B-iv, C-ii, D-i
(3) A-iv, B-iii, C-i, D-ii
(4) A-i, B-ii, C-iii, D-iv

50) Identify the types of cell junctions that help to perform cementing between cells and faciliate communication with neighbouring cells via rapid transfer of ions and small molecules sometimes big molecules respectively:

(1) Tight junctions and Gap junctions, respectively
(2) Gap junctions and Adhering junctions, respectively
(3) Adhering junctions and Gap junctions, respectively
(4) Adhering juctions and Tight junctions, respectively

51) Select the incorrectly matched pair from following:

(1) Osteocytes - one in each lacunae
(2) Chondrocytes - only one in each lacunae
(3) Neurons - Nerve cells
(4) Fibroblast - Areolar tissue

52) Cuboidal epithelium with brush border of microvilli is found in:

(1) Fallopian tube
(2) Proximal convoluted tubule of nephron
(3) Endothelium of blood vessel
(4) Lining of intestine

53) The cells lining the blood vessels belong to the category of:

(1) Connective tissue
(2) Smooth muscle tissue
(3) Squamous epithelium
(4) Columnar epithelium

54) Which of the following is multicellular exocrine gland:-

(1) Goblet cell
(2) Paneth cell
(3) Salivary gland
(4) Both 1 and 2

55) Mast cells secrete:

(1) Histidine
(2) Myoglobin
(3) Heparin
(4) Haemoglobin

56) Four healthy people in their twenties got involved in injuries resulting in damage and death of a few cells of the following. Which of the cells are least likely to be replaced by new cells?

(1) Osteocytes
(2) Epidermis of the skin
(3) Liver cells

(4) Neurons

57) Characteristics of cardiac muscle fibres are:

(1) Cylindrical, branched, striated, uninucleate and involuntary
(2) Spindle-shaped, unbranched, unstriated, multinucleate and involuntary
(3) Cylindrical, unbranched, unstriated, multinucleate and voluntary
(4) Spindle-shaped, unbranched, non-striated, uninucleate and involuntary

58) Inulin is a polymeric polysaccharide consisting of how many types of monosaccharide?

(1) One
(2) Two
(3) Three
(4) Many

59) Exoskeletons of arthropods is made up of:

(1) Chitin
(2) Peptidoglycan
(3) Polymer of N-acetylglucosamine
(4) Both 1 and 3

60) In polysaccharide, the monosaccharides are joined together by:

(1) Ester bonds
(2) Glycosidic bond
(3) Hydrogen bond
(4) Sulphur bond

61) Molecular formula of polysaccharide with 70 glucose unit is

(1) $C_6H_{12}O_6$
(2) $C_{420}H_{840}O_{420}$
(3) $C_{420}H_{602}O_{351}$
(4) $C_{420}H_{702}O_{351}$

62) The bond between 3'-carbon of one sugar of one nucleotide to the 5'-carbon of the sugar of the succeeding nucleotide is a:

(1) Glycosidic bond
(2) Peptide bond
(3) Phosphodiester bond
(4) All of the above

63) Which one is the most appropriate definition for an enzyme?

(1) A protein, acts like a biocatalyst by lowering activation energy in biochemical reactions

(2) A protein, acts like a biocatalyst by increasing activation energy in biochemical reactions

(3) A protein,acts like a biocatalyst by neither lowering nor increasing activation energy in biochemical reactions

(4) A prtein, but not a biocatalyst and has nothing to do with activation energy

64) When nucleic acids that behave like enzymes then that enzyme is called:

(1) Apoenzyme

(2) Coenzyme

(3) Ribozyme

(4) All of these

65) **Statement I :** In a molecule of haemoglobin out of four subunit two of these are identical to each other.
Statement II : Haemoglobin is a tetrameric protein.

(1) Only statement I is correct

(2) Only statement II is correct

(3) Both statements I and II are correct

(4) Both statements I and II are incorrect

66) **Statement I :** Thermostability of all enzymes is in the range of 20°– 35°C.
Statement II : Low temperature preserves the enzyme in at temporarily denatured state.

(1) Both statements I and II are correct

(2) Both statements I and II are incorrect

(3) Only statement I is correct

(4) Only statement II is correct

67) Which enzyme catalyse joining of C-O, C-S, C-N?

(1) Lyase

(2) Hydrolase

(3) Dehydrogenase

(4) Ligase

68) Which of the following statement is correct?

(1) Organic chemists always write a two dimensional view of the molecules

(2) Physicists conjure up the three dimensional views

(3) Biologists describe the protein structure at four levels

(4) All of the above

69) **Statement I :** Lipids are strict biomacromolecules.
Statement II : The macromolecules from cytoplasm and organelles become the acid insoluble

fraction.

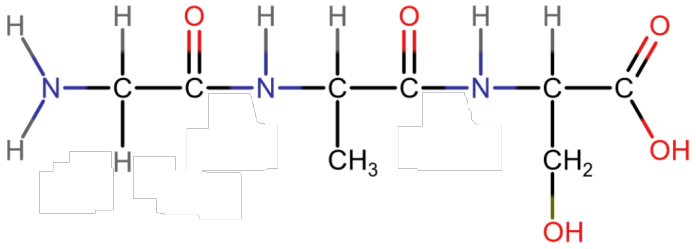
(1) Both statements I and II are correct
(2) Both statements I and II are incorrect
(3) Only statement I is correct
(4) Only statement II is correct

70) Inhibition of succinic dehydrogenase enzyme by malonate is due to:

(1) Malonate cannot bind with succinic dehydrogenase enzyme
(2) Malonate binds with succinate
(3) Malonate binds with succinic dehydrogenase enzyme
(4) Succinate binds with succinic dehydrogenase enzyme

71) Enzymes are composed of one or several polypeptide chains. However, there are a number of cases in which non-protein constituents are bound to the enzyme to make the enzyme catalytically active, in these instances, the protein portion of the enzymes is called:

(1) Holoenzyme
(2) Apoenzyme
(3) Co-factor
(4) All are correct

72) 

(1) Glycine–Serine–Alaline
(2) Glycine–Alaline–Serine
(3) Serine–Alaline–Glycine
(4) Alaline–Serine–Glycine

73) Phospholipid (lecithin) consist of :-

(1) Two fatty acids, one glycerol, one choline phosphate
(2) Two fatty acids, one glyceral, one phosphoric acid
(3) Three fatty acids, one glycerol, one phosphoric acid
(4) One fatty acids, two glycerol, one phosphoric acid

74) The most common sterol is

(1) Phytosterol
(2) Cholesterol

(3) Estradiol

(4) None of these

75) The non-protein moiety of conjugate enzyme is called:

(1) Holoenzyme

(2) Apoenzyme

(3) Co-factor

(4) All are correct

76) For proteolytic enzyme carboxypeptidase which element acts as co-factor:

(1) Fe

(2) Cu

(3) Zn

(4) Co

77) Which given character is true for fatty acid?
(i) A fatty acid has a carboxyl group attached to an R group.
(ii) The R group could be a methyl or ethyl or higher number of $-CH_2$ groups which contain 1 to 19 carbons.
(iii) Fatty acids could be saturated or unsaturated.
(iv) Fats are unsaturated fatty acid.
(v) Fatty acid may be essential or non-essential.

(1) (i), (ii), (iii) and (v)

(2) (i), (iii) and (v)

(3) (i), (iii), (iv) and (v)

(4) All of these

78) Which given character is true for nitrogen bases or nucleic acid found in living organism?
(i) Adenine, guanine, cytosine, uracil and thymine are nitrogen bases.
(ii) When nitrogen base is attached to a sugar, they are called nucleosides.
(iii) If a phosphate group is also found esterified to the sugar of nucleosides then they are called nucleosides.
(iv) Adenosine, guanosine, thymidine, uridine and cytidine are nucleosides.
(v) Adenylic acd, thymidylic acid, guanylic acid, uridylic acid and cytidylic acid are nucleotides.
(vi) Nucleic acids like DNA and RNA consist of nucleotides.

(1) (i), (ii), (iii) and (v)

(2) (i), (ii), (iii), (iv) and (v)

(3) (i), (ii), (iii) and (vi)

(4) All of the above

79) An important quality of such enzymes isolated from thermophilic organisms :-

(1) Optimum temperature for enzymes is 20-35° C

(2) Thermal stability

(3) Damaged at high temperature

(4) Work at very low temperature

80) Low temperature preserves the enzyme in a ...'B'... state whereas high temperature destroys enzymatic activity because proteins are ...'A'... by heat.
Choose the correct option:-

(1) A = temporarily inactive, B = denatured

(2) A = temporarily inactive, B = temporarily inactive

(3) A = permanently inactive, B = denatured

(4) A = denatured, B = temporarily inactive

81)

If temperature is increasing from 30°C to 45°C then the activity of enzyme is :-

(1) First decreases then increases

(2) First increases then decreases

(3) Only increase

(4) Only decrease

82) Which one is correct for competitive inhibitor?
[E = Enzyme, S = Substrate, I = Inhibitor]

(1) ES + I = ESI

(2) S + I = SI

(3) E + I = EI

(4) E + S = ES

83) Consider the following reaction of Glycolysis, Glucose – 6 – Phosphate $\xrightarrow{E}$ Fructose, 6, phosphate, Appropriate code for the enzyme should be :-

(1) 2102

(2) 3431

(3) 5322

(4) 1212

84) In the presence of competitive inhibitor _A_ increases while _B_ remains unchanged :-

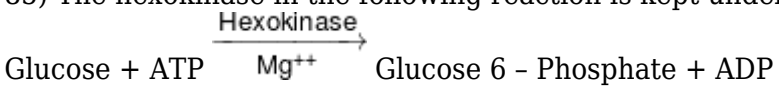
	A	B
(1)	V_{max}	km
(2)	km	V_{max}
(3)	pH	temperature
(4)	temperature	pH

(1) 1

(2) 2

(3) 3
(4) 4

85) The hexokinase in the following reaction is kept under which class ?



(1) Lyase
(2) Ligase
(3) Isomerase
(4) Transferase

86) The biomolecule whose weight is less than___ dalton (Da) is referred to as biomicromolecule :-

(1) 8000
(2) 18
(3) 1000
(4) 10,000

87) Correct order of abundance of oxygen, sulphur, magnesium and calcium in the earth crust ?

(1) O > Ca > Mg > S
(2) S > O > Mg > Ca
(3) Ca > O > S > Mg
(4) Mg > Ca > S > O

88) All monosaccharides can be identified from lipids by Benedict's test, because :-

(1) Presence of chiral carbon, which is present in all monosaccharides
(2) They are found in 'd' & '' forms
(3) They are soluble in water and tasteless
(4) Presence of free aldehyde or ketone group

89) Which of the following Disaccharide is made up of glucose units ?

(1) Cellulose
(2) Starch
(3) Maltose
(4) All the above

90) Which of the following molecule have $\alpha 1 \rightarrow \beta 2$ bond between glucose and fructose?

(1) Chitin
(2) Starch
(3) Inulin
(4) Sucrose

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	3	1	1	1	3	3	2	3	2	4	2	4	3	1	1	2	1	2	1	1
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	4	2	1	3	1	4	1	2	2	2	2	2	1	3	3	2	1	2	2	1
Q.	41	42	43	44	45															
A.	2	1	3	1	2															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	4	3	1	2	2	1	1	3	3	2	2	4	2	3	2	2	4	3	4	1
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	4	3	4	2	4	3	3	4	2	1	4	4	2	4	4	2	3	1	1	4
Q.	86	87	88	89	90															
A.	1	2	3	1	3															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	2	4	1	2	3	1	2	3	4	3	1	1	2	2	2	1	4	3	4	1
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	2	1	3	1	3	3	1	3	4	3	3	2	1	4	4	3	1	3	1	3
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	3	4	1	2	1	2	3	2	3	3	2	2	3	3	3	4	1	1	4	2
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	4	3	1	3	3	2	4	4	4	3	2	2	1	2	3	3	1	4	2	1
Q.	171	172	173	174	175	176	177	178	179	180										
A.	2	3	3	2	4	3	1	4	3	4										

SOLUTIONS

PHYSICS

1) $y = e^x \cdot \cot x$

$\frac{dy}{dx} = e^x \frac{d}{dx}(\cot x) + \cot x \frac{d}{dx}(e^x)$

$= e^x(-\text{cosec}^2 x) + \cot x\, e^x$

$= e^x[\cot x - \text{cosec}^2 x]$

2)

Area of ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$ is 6π

3) Area of disk, $A = \pi r^2$ (where r = radius of disk)

$\frac{dA}{dt} = \pi\left(2r\frac{dr}{dt}\right) = 2\pi r\frac{dr}{dt}$ so $\frac{dA}{dt} = 2\pi \times \frac{5}{\pi} \times 0.1 = 1\ \text{cm}^2/\text{s}$

4)

Use the common integral: $\int \sin(x)dx = -\cos(x)$

$= [-\cos(x)]_0^{\frac{\pi}{3}}$

$= \frac{1}{2}$

5)

Slope of straight line $3y - 6 - \sqrt{3}x = 0$ is $\frac{1}{\sqrt{3}}$

6)

cos 90° = 0 which will make whole product zero.

7)

$y' = 6x^2 - 24x = 0$

x = 0, 4

y" = 12x –24

at x = 4, y" is positive that is minima.

8)

Period of tan = 180°
tan (1837°) = tan (1800 + 37°) = tan 37°

$$\sqrt{101} = (100+1)^{1/2}$$

9) $10\left(1+\frac{1}{100}\right)^{1/2}$

$= 10(1+0.01)^{1/2}$

$\cong 10\left(1+\frac{0.01}{2}\right)$

= 10.05

10) sin 74° = 2 sin 37° cos 37°
= 24/25

tan 74° = $\frac{24}{7}$

11) at x = 0 y = 6
$6 = 2 + ae^0$
a = 4

12) $x = ay^2 + b$

13) $\int_{-1}^{1} e^x dx = [e^x]_{-1}^{1} = e^1 - (e^{-1}) = e - \frac{1}{e}$

14) $y = 2x^3 + 5$

$\frac{dy}{dx} = 2(3x^2) = 6x^2$

Graph between $\frac{dy}{dx}$ and x is a parabola.

15) Substitute u = sin x, then du = cos x dx.

Output: $\int u^2\,du = \frac{u^3}{3} + C$

Back-substitute u = sin x.

Output: $\frac{1}{3}\sin^3 x + C$

16) $[\varepsilon_0] = \frac{[Q]^2}{[F][r]^2}$

$[\mu_0] = \frac{[F][d]}{[\ell][i]^2}$

$$[E] = \frac{[F]}{[Q]}[B] = \frac{[F]}{[i][L]}$$

17)

$n_1u_1 = n_2u_2$

$$n_1 = \frac{n_2u_2}{u_1} = \left(\frac{M_2}{M_1}\right)^1\left(\frac{L_2}{L_1}\right)^1\left(\frac{T_2}{T_1}\right)^{-2}$$

$$= \left(\frac{10gm}{1000gm}\right)\left(\frac{10cm}{100cm}\right)\left(\frac{0.1}{1}\right)^{-2}$$

$$= \frac{1}{100} \times \frac{1}{10} \times 0.1 \times 0.1$$

$$= \frac{1}{10} = 0.1\ N$$

18) $N_1U_1 = N_2U_2$.

19)

$T = P^a D^b S^c$

$\Rightarrow \quad [M^0 L^0 T^1] = [M L^{-1} T^{-2}]^a[ML^{-3}]^b[MT^{-2}]^c$

$= M^{a+b+c} L^{-a-3b} T^{-2a-2c}$

$\Rightarrow \quad a + b + c = 0$

$-a - 3b = 0$

$-2a - 2c = 1$

$$\Rightarrow \quad a + c = -\frac{1}{\sqrt{2}}$$

$$b = \frac{1}{2},\ a = \frac{-3}{2}\ , c = 1$$

20) $\left[\frac{\alpha z}{k\theta}\right]$ dimensionless

$$\alpha = \frac{k\theta}{z} \Rightarrow [\alpha] = \frac{[ML^2T^{-2}k^{-1}][k]}{[L]}$$

$$a = [MLT^{-2}] \quad \text{and } P = \frac{\alpha}{\beta}$$

$$\Rightarrow \ [\beta] = \frac{[\alpha]}{[p]} = \frac{[MLT^{-2}]}{[ML^{-1}T^{-2}]} = [M^0 L^2 T^0]$$

21) $R = \sqrt{AB}$

$R = r \pm \Delta r$

where, $r = \sqrt{ab}$

and $$\frac{\Delta r}{r} = \frac{1}{2}\frac{\Delta a}{a} + \frac{1}{2}\frac{\Delta b}{b}$$

22)

As $P = \frac{a^3b^2}{cd}$

$\frac{\Delta P}{P} = \frac{3\Delta a}{a} + 2\frac{\Delta b}{b} + \frac{\Delta c}{c} + \frac{\Delta d}{d}$

so % error in P is

$\frac{\Delta p}{p} \times 100$ = (3 × 1 + 2 × 2 + 3 + 4)%

= 14%

23)

No. of observation $\propto \frac{1}{\text{Error}}$

24) $\frac{\Delta V}{V} = 3 \times \frac{\Delta r}{r} = 3 \times \frac{1}{100} = \frac{3}{100} = 3\%.$

25) $R_1 = (100 \pm 3)\Omega,\ R_2 = (200 \pm 4)\Omega$

If R_P is result of parallel combination then

$\frac{1}{R_P} = \frac{1}{R_1} + \frac{1}{R_2}$ ...(1)

$\Rightarrow R_P = \frac{200}{3} = 66.7\Omega$

on differentiating equation (1)

$\frac{dR_P}{R_P^2} = \frac{dR_1}{R_1^2} + \frac{dR_2}{R_2^2}$

$dR_P = R_P^2\left(\frac{dR_1}{R_1^2} + \frac{dR_2}{R_2^2}\right)$

$dR_P = \left(\frac{200}{3}\right)^2\left(\frac{3}{100^2} + \frac{4}{(200)^2}\right)$

$= \frac{16}{9} \approx 1.8$

In the form of

$R_P = R_P \pm dR_P$

$R_P = \left(\frac{200}{3} \pm \frac{16}{9}\right)\Omega$

or $R_P = (66.7 \pm 1.8)\Omega$

26) $P = \frac{a^4b^3}{c^2d}$

$\frac{dp}{P} \times 100 = 4\frac{da}{a} \times 100 + 3\frac{db}{b} \times 100 + 2\frac{dc}{c} \times 100 + \frac{dd}{d} \times 100$

% Error in P = 4 × 0.025 + 3 × 0.5 + 2 ×1+1.5

= 6%

27) Both sides should be dimensionless

28) Theory Based

29)

From rule of order of magnitude
$A = 640 = 0.640\times10^{3} \Rightarrow x = 3$
$B = 8.2\times10^{4} = 0.82\times10^{5} \Rightarrow x = 5$
$C = 120 \times10^{2} = 1.2\times10^{4} \Rightarrow x = 4$
$D = 72000 \times10^{-1} = 0.72\times10^{4} \Rightarrow x = 4$

30) Theory based

31) L = 10.00cm ± 0.01 cm
R = 1.000 cm ± 0.001 cm

$$\Delta L\% = \frac{0.01}{10.00} \times 100 = 0.1\%$$

$$\Delta R\% = \frac{0.001}{1.000} \times 100 = 0.1\%$$

$v = \pi R^2 L$
$\Delta v\% = 2\Delta R\% + \Delta L\% = 2(0.1\%) + 0.1\% = 0.3\%$

32)

Since measured length is 3.50 cm ⇒ Least count of measurement = 0.01 cm
(1) L.C. of meter scale is 1mm = 0.1 cm
(2) L.C of vernier caliper = M.S.D-VSD (as 10VSD = 9MSD)

$$\text{L.C} = \text{MSD} - \frac{9}{10}\text{MSD} = \frac{1}{10}(\text{MSD}) = 0.01 \text{ cm}$$

(3) L.C of screw gauge (100 division) $= \frac{\text{Pitch}}{\text{division}} = \frac{1\text{mm}}{100} = 0.01$ mm $= 0.001$ cm

(4) L.C of screw gauge (50 division) $= \frac{\text{Pitch}}{\text{division}} = \frac{1\text{mm}}{50} = 0.02$ mm $= 0.002$ cm

33) When jaws are closed, the zero error will be = main scale reading + (circular scale reading) (Least count)
= 0 mm + (45)(0.01)
Negative zero error = –0.05 mm
When the sheet is placed between the jaws; measured thickness
= 0.5mm + (25)(0.01) = 0.75mm
⇒ Actual thickness
= 0.75mm – (–0.05) mm
= 0.80mm

34) Pitch p = $\frac{1\text{mm}}{2} = 0.5\text{mm}$ & least count (LC) = $\frac{\text{pitch}}{\text{No. of CSD}} = \frac{0.5\text{mm}}{100}$ = 0.005 mm

In fig (1) No measurement : There is a Negative zero error

Ze = –4 CSD = – 4 × 0.005 mm = – 0.02 mm

In fig (1) During measurement : MSR = 4 pitch = 4 × 0.5 mm = 2 mm

CSR = 55 CSD = 56 × 0.005 mm = 0.280 mm

Final reading = 2 mm + 0.280 mm –(–0.2 mm) = 2.300 mm

35) Accuracy is closeness of a measured value to actual value, which is independent of measuring instrument's least count.

36) 50 divisions = 2.45 cm

$\Rightarrow$ 1 division = $\frac{2.45}{50} = 0.049\text{cm}$

$\Rightarrow$ least count = 1 MSD –1VSD = 0.05 – 0.049

= 0.001 cm.

So vernier reading = 0.001 × 24 = 0.024 cm.

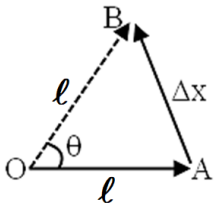
Therefore diameter of cylinder

= 5.10 + 0.024 = 5.124 cm.

37) $F_{net} = GMm\left[\frac{1}{r^2} + \frac{1}{2r^2} + \frac{1}{4r^2} + ...\infty\right]$

$= GMm\left[\frac{1/r^2}{1-\frac{1}{2}}\right] = \frac{2GMm}{r^2}$

38)



Δx = change in position vector of head

$= \left|\overrightarrow{OB} - \overrightarrow{OA}\right|$

$= 2\ell\sin\frac{\theta}{2}$

39) $\vec{C} = x\hat{i} + y\hat{j} + z\hat{k}$

If $\vec{C}$ is $\perp$ to $\vec{A}$ and $\vec{B}$ then

$\left.\begin{array}{l}\vec{C}.\vec{A} = 0 \Rightarrow x - 2y + z = 0 \\ \vec{C}.\vec{B} = 0 \Rightarrow x + 2y = 0\end{array}\right\} 2x + z = 0$

and $\left|\vec{C}\right| = 1 = \sqrt{x^2 + y^2 + z^2} = 1$

$x^2 + y^2 + z^2 = 1$

$x^2 + \frac{x^2}{4} + 4x^2 = 1 \Rightarrow \frac{21x^2}{4} = 1$

$\Rightarrow x = -\frac{2}{\sqrt{21}}$

$z = \frac{4}{\sqrt{21}}, \; y = \frac{1}{\sqrt{21}}$

40)

$F_{max} = F_1 + F_2 = 10$ N

$F_{min} = F_1 - F_2 = 6$ N

So $F_1 = 8$ N

$F_2 = 2$ N

and force is increased by 3 N

$= F_1^1 = 11$ N, $F_2^1 = 5$ N

Now

$R = \sqrt{10^2 + 5^2 + 2 \times 11 \times 5 \cos 90^\circ}$

$= \sqrt{146}$ N

41) $\vec{R} = \vec{A} + \vec{B}$

$\vec{R} = 6\hat{i} - \hat{j} - 3\hat{k}$

$\hat{R} = \frac{\vec{R}}{|R|} = \frac{6\hat{i} - \hat{j} - 3\hat{k}}{\sqrt{36 + 1 + 9}} = \frac{1}{\sqrt{46}}\left(6\hat{i} - \hat{j} - 3\hat{k}\right)$

42)

$\left|\vec{A}_1\right| = 2 \quad \left|\vec{A}_2\right| = 3 \quad \left|\vec{A}_1 + \vec{A}_2\right| = 3$

$\left|\vec{A}_1 + \vec{A}_2\right| = \sqrt{A_1^2 + A_2^2 + 2A_1A_2\cos\theta}$

$3 = \sqrt{4 + 9 + 2A_1A_2\cos\theta}$

$9 = 4 + 9 + 2A_1A_2 \cos\theta$

$\boxed{2A_1A_2\cos\theta = -4}$

$\left(\vec{A}_1 + 2\vec{A}_2\right)\left(3\vec{A}_1 - 4\vec{A}_2\right)$

$3A_1^2 - 4A_1A_2\cos\theta + 6A_1A_2\cos\theta - 8A_2^2$

$12 + 2A_1A_2 \cos\theta - 72$

$12 - 4 - 72$

$\Rightarrow 8 - 72$

$\Rightarrow -64$

43) Vector component of $\vec{A}$ along $\vec{B}$ is

$\left\{\left|\vec{A}\right| \cos\theta\right\}\hat{B} = \left(\vec{A}.\hat{B}\right)\hat{B}$

$\hat{B} = \frac{\vec{B}}{\left|\vec{B}\right|} = \frac{i + j + k}{\sqrt{1 + 1 + 1}} = \frac{i + j + k}{\sqrt{3}}$

$\vec{A}\hat{B} = (3i + 4j + 5k) = \left(\frac{i + j + k}{\sqrt{3}}\right)$

44) $\vec{A} = \hat{i} + \hat{j} + \hat{k}, \vec{B} = \hat{i} + \hat{j}$

$A = \sqrt{1^2 + 1^2 + 1^2} = \sqrt{3}, B = \sqrt{1^2 + 1^2} = \sqrt{2}$

$\vec{A} \cdot \vec{B} = 2$

$\cos\theta = \frac{\vec{A}\vec{B}}{AB} = \frac{2}{\sqrt{3}\sqrt{2}} = \left(\sqrt{\frac{2}{3}}\right)$

$\sin\theta = \sqrt{1 - -\cos^2\theta} = \sqrt{1\frac{2}{3}} = \sqrt{\frac{1}{3}} = \frac{1}{\sqrt{3}}$

$\therefore \quad \theta = \sin^{-1}\left(\frac{1}{\sqrt{3}}\right)$

45) $y_{avg} = \frac{\int y dx}{\int dx} = \frac{\text{area}}{3} \Rightarrow \frac{\frac{1}{2}(3 + 2)\,20}{3} \Rightarrow \frac{50}{3}$

CHEMISTRY

46) ${}_{1}^{1}H$ has no neutron

47)

Schrodinger equation give 3 quantum numbers, n, l and m.

48)

The third energy level of an atom contains a total of 9 atomic orbitals. This includes one s orbital, three p orbitals, and five d orbitals.

49)

$$\text{mole (n)} = \frac{\text{Weight}}{\text{Molecular weight}} = \frac{\text{Volume (at STP)}}{22.4}$$

$$\frac{1.25}{M} = \frac{1}{22.4}$$

Molecular weight = 1.25 × 22.4 = 28

In given option molecular weight of CO = 28

50)

	CH_4	:	SO_3
Mass	2	:	5
Mole	$\frac{2}{16}$	:	$\frac{5}{80}$

or 2 : 1

51) No. of molecules = $\frac{\text{wt.}}{\text{mol. wt.}} \times N_A$

$= \frac{0.2}{44} \times 6.023 \times 10^{23}$

$= 2.74 \times 10^{21}$ molecules

The residual CO_2 molecules after removal of 10^{21} molecules

$= 2.74 \times 10^{21} - 10^{21}$

$= 1.74 \times 10^{21}$ molecules

No. of moles = $\frac{\text{No. of molecules}}{N_A}$

$= \frac{1.74 \times 10^{21}}{6.023 \times 10^{23}} = 2.88 \times 10^{-3}$

52) $d = \frac{m}{v} = \frac{16\ g}{22.4\ L}$

53) V.D. = 36 ⇒ MM = 72

⇒ 12n + (2n + 2) = 72

⇒14n = 70 ⇒ n = 5

54)

At same temperature, pressure, volume different gases will have same no. of mole of gases.

∴ Suppose 1 mole of each gas occupies

$1H_2 \rightarrow N_A \times 2\,\text{atoms}$

$1He \rightarrow 1\,\text{atom} \times N_A$

$1\ O_2 \rightarrow 2\,\text{atom} \times N_A$

$O_3 \rightarrow 3\,\text{atom} \times N_A$

∴ Ratio will be 2 : 1 : 2 : 3

55) **Correct : Statement I :** In ground state configuration of $_{24}Cr$ number of electrons with n + $l \geq 3$ is 20.

Correct : Statement II : In $_{36}Kr$ total number of electrons with m = 0 and s = +1/2 is 8.

56) ln 159.5 g $CuSO_4$ = 63.5 g Cu

1g $CuSO_4$ = $\frac{63.5}{159.5}$g Cu

100g $CuSO_4$ = $\frac{100 \times 63.5}{159.5}$g Cu

$= 39.8 \simeq 40g$

57) Butane and isobutane have same molecular formula, thus

$C_4H_{10} + 6.5\ O_2 \rightarrow 4\ CO_2 + 5\ H_2O$

58 g C_4H_{10} requires O_2 = 6.5 × 32 g

∴ 1000 g C_4H_{10} requires O_2 = $\frac{6.5 \times 32 \times 1000}{58}$ = 3.586 Kg.

58)

$$C_4H_{10(g)} + \frac{13}{2} O_{2(g)} \rightarrow 4CO_{2(g)} + 5H_2O_{(g)}$$

100 mL	400 mL	500 mL
200 mL	800 mL	1000 mL

59)

C	H
$\frac{80}{12}$	$\frac{20}{1}$

2 × V.D = M_w

60) $2KClO_3 \xrightarrow{\Delta} 2KCl + 3O_2(g)$

2mole of $KClO_3$ gives

3 mole of O_2

For ½ mole of O_2

0.33 mole of $KClO_3$ will be required mole $KClO_3$ = $\frac{2}{3} \times \frac{1}{2} = \frac{1}{3} = 0.33$

61) **Explanation :**

The question involves a neutralization reaction between a tribasic acid and sodium hydroxide (NaOH). We are given the mass of NaOH and the mass of the tribasic acid required for complete neutralization, and we need to determine the molar mass of the tribasic acid.

Given Data :

Mass of NaOH = 12 g

Mass of tribasic acid = 9.8 g

Concept :

Neutralization Reaction : A neutralization reaction is a chemical reaction between an acid and a base, typically resulting in the formation of salf and water.

Stoichiometry : The balanced chemical equation for the reaction provides the mole ratios between the reactants. In this case, since the acid is tribasic, one mole of the acid will react with three moles of NaOH

Molar Mass : The molar mass of a substance is the mass of one mole of that substance.

Calculate moles of NaOH :

Molar mass of NaOH = 40 g/mol

Moles of NaOH = Mass of NaOH / Molar mass of NaOH

Moles of NaOH = 12g/40g/mol = 0.3 moles

Calculate moles of tribasic acid :

Since one mole of tribasic acid reacts with three moles of NaOH :

Moles of trivasic acid = Moles of NaOH/3

Moles of tribasic acid = 0.3 moles / 3 = 0.1 moles

Calculate molar mass of tribasic acid :
Molar mass of tribasic acid = Mass of tribasic acid / Moles of tribasic acid
Molar mass of rribasic acid = 9.8 g/ 0.1 moles = 98 g/mol

62) $3CaBr_2 + 2K_3PO_4 \rightarrow Ca_3(PO_4)_2 + 6KBr$

t = 0	0.5 mol	0.2 mol (L.R.)	0	0
After reaction	0.5 – 0.3 = 0.2 mol	0	0.1 mol	0.6 mol

63)

The balanced chemical reaction is $\frac{1}{2}O_2(g) + H_2(g) \rightarrow H_2O$
Initially, say V mL of oxygen was present.
The volume of hydrogen present initially was 4 mL
After reaction, V_2 mL of oxygen and 0 mol of hydrogen are present.
However, 8 mL of oxygen remained after the reaction.
Thus, V_2 = 8 mL or, V = 10 mL

64) Balanced equation will be $N_2 + 3H_2 \rightarrow 2NH_3$
Given: 100mL 100 mL
For determination of Limiting reagent divide the given quantities by stoichiometric coefficients
$\frac{100}{1} = 100$ $\frac{100}{3} = 33.3$ (Limiting reagent)
In this reaction, H_2 is limiting reagent so reaction will proceed according to H_2.
According to stoichiometry from 3 mL of H_2, produced volume of NH_3 = 2 mL
That is, from 100 mL of H_2, produced volume of $NH_3 = \frac{2}{3} \times 100 = 66.6$ mL

65) (A) Equivalents of $H_2C_2O_4.2H_2O$ in 10 mL = Equivalents of NaOH
$\left(\frac{6.3}{126/2} \times \frac{1000}{250}\right) \times \frac{10}{1000} = 0.1 \times V$ (in litre)
 V = 0.04 L = 40 mL.

66)

When two element combine with each other to form more than one compound.
Here, Sn and Cl combine with each other to form $SnCl_2$ and $SnCl_4$.

67) I oxide = MO
M : O
mol : $\frac{77.78}{M} : \frac{22.22}{16} = 1:1$
⇒ M = 56
II oxide = ?

M : O

$\frac{70}{56} : \frac{30}{16}$

mol : = 2 : 3 $\Rightarrow M_2O_3$

68)

Question Explanation:
This question asks us to identify the statement that is incorrect regarding the properties of cathode rays.

Concept:

A. **Cathode rays**

Analysis:

A. **Statement (1):** Cathode rays are indeed a stream of electrons. This is a fundamental property.
B. **Statement (2):** Electrons are charged particles, so this is also true.
C. **Statement (3):** Moving charged particles are deflected by magnetic fields, so this is also true.
D. **Statement (4):** While cathode rays move at high speeds, they do not move at the same speed as light. Their speed is typically a fraction of the speed of light.

Final Answer:
The statement that is not true with respect to cathode rays is that they move with the same speed as that of light.
Therefore the correct option is (4).

69) $E = \frac{nhc}{\lambda}$

$p \times t = \frac{nhc}{\lambda}$

$\Rightarrow \frac{n}{t} = \frac{p\lambda}{hc} = \frac{100 \times 400 \times 10^{-10}}{3 \times 10^8 \times 6.6 \times 10^{-34}}$

$\frac{n}{t} = \frac{4 \times 10^{20}}{19.8} = 2 \times 10^{19}$

70)

Incorrect : Statement I : According to Heisenberg's product of uncertainty in measurement of velocity and uncertainty in measurement of position is $\geq h/4\pi$.
Correct : Statement II : $Fe_2O_3 + Al \rightarrow Fe + Al_2O_3 + Q$ ratio of masses of Fe_2O_3 and Al taken to given maximum energy is 80 : 27.

71) n – □ – 1 = 1

$\Rightarrow$ 3 – □ – 1 = 1

$\Rightarrow$ 2 – □ = 1

$\Rightarrow$ □ = 1

orbital angular momentum = $\frac{h}{2\pi}\sqrt{\ell(\ell+1)}$

$= \frac{h}{2\pi}\sqrt{1\times 2}$

$= \sqrt{2}\frac{h}{2\pi}$

72) **Incorrect : Statement I :** On diluting solution, it's normality and molarity changes but molality remains constant.
Incorrect : Statement II : The total number of atoms in 4 amu He and 36 amu H_2O will be $7N_A$.

73) $E = hC\bar{\nu}$

□ $\frac{E_1}{E_2} = \frac{3}{8} = \frac{150}{\bar{\nu}_2}$; $\bar{\nu}_2 = 400 cm^{-1}$

74)

Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.

75) $r_n = 0.529 \times \frac{n^2}{Z}$ Å

means, $r_n \propto \frac{1}{Z}$

76) In H-atom ionisation energy is less than 1 eV is for n = 4 (E_4 = –0.85 eV) on wards. therefore minimum value of n = 4.

77) **Question Explanation :**
Identify which statement about quantum numbers and orbitals is incorrect.

Concept :
Energy levels and quantum numbers vary depending on subshell and atom type; in hydrogen atom, energy depends only on principal quantum number n.

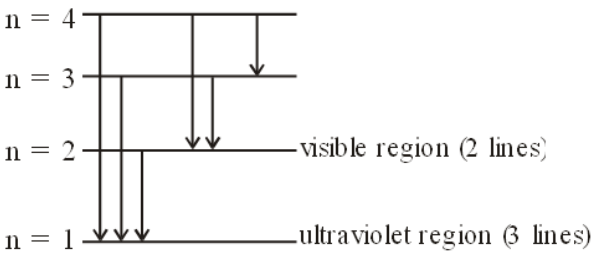
Solution :
• Statement I : Azimuthal quantum number l for d-subshell is 2 regardless of principal quantum number n, so it's same for 3d, 4d, 5d – correct.
• Statement 2 : Electron distribution in d_{xy} orbital lies between axes, while p_z orbital is along the x-axis - Correct.
• Statement 3 : ϕ520 corresponds to $5d_{x^2}$ orbital – Correct (hear 5 = n- l, 0=m_l)
• Statement 4 : In hydrogen atom, energy depends only on n, so energy of 5d (n=5) and 4f (n =4) are not the same – Incorrect statement.

Final Answer :

Option (4)

78) **Explanation :**
The question asks for the transition in a He^+ ion that will emit the maximum energy radiation. The energy emitted depends on the difference in energy level between the initial and final states. A larger energy difference will result in the emission of higher energy radiation.

Given :
The energy level of a hydrogen-like ion (such as He^+) are given by the formula
$E_n = -\frac{13.6}{4n^2}eV$
Concept :
Energy difference for Transition
Calculation :
We calculate the energy levels for each transitions and find the energy differences
• **Transition from** n = 5 to n = 2
$E_5 = -\frac{13.6}{4\times5^2} = -0.136eV, E_2 = -\frac{13.6}{4\times2^2} = -0.85eV$
Energy difference : $\Delta E = E_2 - E_5$ = –0.85 – (–0.136) = 0.714 eV
• **Transition from** n = 3 to n = 1
$E_3 = -\frac{13.6}{4\times3^2} = -0.337eV, E_1 = -\frac{13.6}{4\times1^2} = -3.4eV$
Energy difference :
$\Delta E = E_1 - E_3$ = –3.4 – (–0.377) = 3.023 eV
• **Transition from** n = 7 to n = 3
$E_7 = -\frac{13.6}{4\times7^2} = -0.069eV, E_3 = -0.377eV$
Energy difference : $\Delta E = E_3 - E_7$ = –0.377 – (–0.069) = 0.308 eV
• **Transition from** n = 8 to n = 4
$E_8 = -\frac{13.6}{4\times8^2} = -0.053eV, E_4 = -0.2125eV$
Energy difference :
$\Delta E = E_4 - E_8$ = –0.2125 – (–0.053) = 0.1595 eV

79) 

80) $\frac{1}{\lambda} = R_H \times Z^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$

81)

Generated by Allie

Problem Statement: The question asks us to find the longest wavelength in the Paschen series of the lithium ion Li^{2+}, given that the shortest wavelength of the helium ion He^{+} in the Balmer series is x. We need to compare these wavelengths using the appropriate spectral series formulas for hydrogen-like ions.

Underlying Concept: The problem involves the concept of spectral lines of hydrogen-like ions, which are described by the Rydberg formula. Each spectral series (Balmer, Paschen, etc.) corresponds to electronic transitions ending at a specific energy level (n_1). The wavelength emitted depends on the difference in energy levels (n_2 to n_1) and the nuclear charge Z^2, where Z is the atomic number of the ion.

Relevant Formulas: The Rydberg formula used is:

$$\frac{1}{\lambda} = RZ^2\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$$

where: - λ is the wavelength of emitted light, - R is the Rydberg constant for hydrogen, - Z is the atomic number of the ion, - n_1 is the lower energy level (fixed for each series), - $n_2 > n_1$ is the higher energy level involved in the transition.

Step-by-Step Calculation: 1. For He^{+} (Z=2) in the Balmer series (n_1=2), the shortest wavelength corresponds to the transition from n_2 â†’ âˆž:

$$\frac{1}{\lambda_{min}} = R \times 2^2 \times \left(\frac{1}{2^2} - 0\right) = R \times 4 \times \frac{1}{4} = R$$

This gives $\lambda_{min} = \frac{1}{R} = x$.

2. For Li^{2+} (Z=3) in the Paschen series (n_1=3), the longest wavelength corresponds to the smallest energy difference, i.e., transition from n_2=4 to n_1=3:

$$\frac{1}{\lambda_{max}} = R \times 3^2 \times \left(\frac{1}{3^2} - \frac{1}{4^2}\right) = 9R \times \left(\frac{1}{9} - \frac{1}{16}\right) = 9R \times \frac{16-9}{144} = 9R \times \frac{7}{144} = \frac{7}{16}R$$

Therefore,

$$\lambda_{max} = \frac{16}{7R} = \frac{16}{7}x$$

.

Hence, the longest wavelength in the Paschen series of Li^{2+} is $\frac{16x}{7}$.

Tips and Tricks: Use the fact that for hydrogen-like ions, the energy transitions depend on Z^2, which scales the Rydberg constant. Also recall that spectral series are defined by fixed lower levels for n_1.

Common Mistakes: Students sometimes confuse the shortest and longest wavelengths with the same transition distances or forget to square the atomic number Z when calculating wavelengths for hydrogen-like ions. Another common error is mixing up n_1 and n_2 values for the series.

82) $\lambda = \frac{h}{mv} = \frac{6.6 \times 10^{-34}}{66 \times 10^3}$

83) ∵ $2\pi r = 2\lambda \quad 2\pi r = 2x \quad r = \frac{x}{\pi}$

84) $\Delta V = \frac{h}{4\pi m \Delta x}$

m = 0.01 kg, Δx = 10^{-5} m h = 6.63 × 10^{-34} Js

85)

$Fe^{+2} = 1s^2 2s^2 2p^6 3s^6 3d^6$

86)

Generated by Allie
Problem Statement: The question asks where the probability of finding an electron in a pâ„“ atomic orbital is zero. In simpler terms, it wants to know on which plane or direction the electron cannot be found if it is in the pâ„“ orbital.
Underlying Concept: The problem involves the shape and spatial orientation of atomic orbitals, specifically the pâ„“ orbital. The p orbitals have directional dependent wavefunctions, and the electron probability density depends on the value of the wavefunction. If the wavefunction is zero at some locations, the probability of finding the electron there is zero.
Relevant Formulas: The wavefunction for a pâ„“ orbital can be represented as:

$$\Psi \propto x e^{-r/a}$$

where: - x is the coordinate along the x-axis, - r is the radial distance, - a is a constant related to the atomic orbital size.
Since the probability density is proportional to $|\Psi|^2$, if $\Psi = 0$ at any point, the probability of finding the electron there is zero.
Step-by-Step Calculation: The wavefunction for a pâ„“ orbital involves the factor x. On the yz-plane, $x = 0$. Plugging this into the wavefunction: $\Psi = x e^{-r/a} = 0$ Thus, the probability density $|\Psi|^2 = 0$ in the yz-plane, meaning the electron cannot be found there.
Tips and Tricks: Remember that p orbitals have a nodal plane where the probability of finding an electron is zero. For the pâ„“ orbital, this nodal plane is the yz-plane.
Common Mistakes: Confusing which plane defines zero probability for p orbitals. For example, thinking the zero probability is in the xy or xz plane instead of the correct yz plane for the pâ„“ orbital.
Why Other Options Are Incorrect?: â€¢ 'In the xy plane' or 'in the xz plane' are incorrect because $x \neq 0$ on these planes; hence the wavefunction is not zero. â€¢ 'In the x-direction' is incorrect because the probability is zero on an entire plane, not just along a direction.

87)

Explanation

A. n + □ rule: This rule helps determine the energy levels of electrons in an atom. 'n' is the principal quantum number, and '□' is the azimuthal quantum number.

Concept

A. n + □ = 5: We need to find the combinations of n and □ that satisfy this condition:

    A. n = 5, □ = 0 (5s)

B. n = 4, □ = 1 (4p)

C. n = 3, □ = 2 (3d)

B. Orbitals:

A. 5s: 1 orbital (2 electrons max)

B. 4p: 3 orbitals (6 electrons max)

C. 3d: 5 orbitals (10 electrons max)

C. Electrons of same spin:

A. 5s : 1 electron with same spin

B. 4p : 3 electrons with same spin

C. 3d : 5 electrons with same spin

D. Total number of electrons with same spin = 1 + 3 + 5 = 9

Answer option 2, (9).

88)

Explanation

A. Pauli's exclusion principle states that no two electrons in an atom can have the same set of four quantum numbers (i.e., no two electrons can have the same spin in the same orbital). Hund's rule states that electrons will singly occupy each orbital within a subshell before any orbital is doubly occupied, and all electrons in singly occupied orbitals will have the same spin.

Concept

A. A violation of both rules means an orbital has two electrons with the same spin AND some orbitals in the subshell are empty while others are doubly occupied.

B. An orbital diagram showing two electrons with the same spin in an orbital AND some empty orbitals in the same subshell while another orbital is doubly occupied.

Answer option 3

89) $hn - hn_0 = eV_S$

$v = v_0 + \frac{e}{h}V_S$

90) Conceptual

BIOLOGY

91) Number of chromosomes remain same in mitosis. The gametes are formed by reductional division i.e. meiosis.

92) **Explaining :**
Concept :
Solution : Meiosis occurs in **germ cells**, not somatic cells.
Mitosis, not meiosis, is responsible for growth and repair.
Final Answer : (4)

93) NCERT-XI, Page # 126
Both Assertion and Reason are true, and Reason is the correct explanation of Assertion.

94) NCERT-XI, Page # 126
During leptotene stage the chromosomes become gradually visible under the light microscope.
The compaction of chromosomes continues throughout leptotene.
The beginning of diplotene is recognised by the dissolution of the synaptonemal complex.

95)

NCERT Pg.#123,125

96) NCERT XI, Page # 121

97)

NCERT-XI, Pg. # 128

98)

NCERT-XI, Pg. # 121

99)

NCERT-XI, Pg. # 124

100)

NCERT-XI, Page # 127

101) The assertion is true as during anaphase II of meiosis II, the sister chromatids do indeed separate and move to opposite poles. The reason is also true and is the correct explanation for the assertion because the centromeres holding the sister chromatids together split, and the microtubules attached to the kinetochores shorten, which facilitates the movement of the chromatids to the poles.

102) The assertion is true as chiasmata, which are X-shaped structures, represent the sites of crossing over and are visible during the diplotene stage of meiosis. The reason is also true and is the correct explanation for the assertion because the synaptonemal complex, which holds homologous chromosomes together, dissolves at the beginning of diplotene, allowing the chromosomes to separate but remain connected at the chiasmata where crossing over has occurred.

103) The assertion is true as mitotic divisions in meristematic tissues, including the apical and lateral cambium, enable plants to grow continuously throughout their life. The reason is also true and is the correct explanation for the assertion because cell division through mitosis helps in maintaining the nucleo-cytoplasmic ratio, which gets disturbed during cell growth. This is essential for the proper functioning of cells and the overall growth of the plant.

104) Statement I is true as during metaphase, chromosomes align at the cell's equator, and each of their kinetochores is indeed attached to spindle fibres coming from opposite poles. statement II is also true because the metaphase plate is the specific plane where chromosomes line up during metaphase, which is facilitated by the attachment of spindle fibres to their kinetochores.

105) The assertion is true as it correctly states that mitotic cell division in animals generally occurs in diploid somatic cells with some exceptions like the haploid cells of male honey bees. The reason is also true and correctly explains the assertion by stating that plants have the ability to undergo mitotic divisions in both haploid and diploid cells, which is a contrast to the general pattern seen in animals. Therefore, both A and R are true, and R is the correct explanation of A.

106) New NCERT Page no. 125

107) NCERT-XI, Pg. # 121

108) NCERT-XI, Pg. # 127

109) NCERT XI, Page # 126

110) NCERT-XI, Page No. 165, 166, 169

111) Ncert page 99.

112) NCERT, Pg. # 93

113)

NCERT XI Pg. # 90, 91

114)

NCERT XI Pg. # 90

115) NCERT XI Pg. No. # 88

116) NCERT Pg. No. 95, 96

117) NCERT Pg. No. 94

118) NCERT-XI, Pg. # 94, 95

119) **The correct answer is: 4.** They form pellicle
Fimbriae are small, hair-like structures on the surface of bacteria that help in attachment to surfaces, including host tissues or other surfaces like rocks in streams. However, they do not form a pellicle. A pellicle is a thin layer formed by other types of structures, typically associated with biofilms or certain algae.

120) NCERT -XI, Pg. # 94

121) NCERT Page no. 96

122)

NCERT Pg.No. 96

123)

NCERT Pg.No. 101, 102

124)

NCERT Pg.No. 96

125) NCERT-XI, Pg. # 100

126) NCERT pg. 138 2021-22

127) NCERT-XI, Pg. # 98

128) NCERT pg no.128,132,138,139

129)

NCERT Pg. # 136

130) NCERT Page 87.

131)

Explaining the Question – Assertion and reason type question with the reference of vacuole, two statements are given in question first statement is assertion and other is reason.
Concept – This Question is based on Centrioles and Nucleus.
Solution –

Assertion: Centriole form the basal body of cilia or flagella. True

Reason: Nucleolus is a site for inactive RNA synthesis.-False

True is - Nucleolus is a site for active ribosomal RNA synthesis.
Final Answer - 3

132) Statement I: Correct. The extensions such as vesicles, tubules, and lamellae in cells indeed help in various cellular processes including cell wall formation, DNA replication, and distribution to daughter cells.
Statement II: Incorrect. Chromatophores are membranous extensions found in some prokaryotes like cyanobacteria that contain pigments and are involved in photosynthesis, not in motility.

133) Statement I is true as the axoneme of cilia and flagella indeed has a 9+2 pattern of microtubules, which includes nine doublets of radially arranged peripheral microtubules and a pair of centrally located microtubules. Statement II is true, as the 9+2 arrangement specifically describes this unique structure of microtubules within the axoneme.

134) The assertion is true as chromoplasts do contain carotenoid pigments which give plant parts their characteristic colors. The reason is also true as leucoplasts are indeed colorless plastids that store various nutrients. However, the reason R is not the correct explanation of A

because it does not explain why chromoplasts contain carotenoid pigments; instead, it describes a different function of another type of plastid, the leucoplast.

135) Statement I is true as Reductionist Biology does focus on understanding the molecular basis of physiological processes. Statement II is also true because Reductionist Biology indeed applies the concepts and techniques of physics and chemistry to understand biological processes, which is the essence of the physico-chemical approach mentioned in the text.

136) **Incorrect : Statement I :** Ligaments are dense irregular tissue.
Incorrect : Statement II : Cartilage is dense regular tissue.

137)

Conceptual

138)

Option 2 is correct.

139)

Conceptual

140)

Option 3 is correct Adhering junctions and Gap junctions, respectively

141)

Chondrocytes - only one in each lacunae" because multiple chondrocytes can reside in a single lacuna within cartilage tissue.

142)

Cuboidal epithelium with a brush border of microvilli is primarily found in the proximal convoluted tubule (PCT) of the nephron in the kidney.

143)

The cells lining blood vessels belong to the category of simple squamous epithelium.

144)

It seems like the answer options are missing from your query. Examples of multicellular exocrine glands include sweat glands, salivary glands, mammary glands, and liver.

145)

Mast cells occur in connective tissue proper, these are modified basophills of blood. Mast cells secrete heparin histamine and serotonin.

146)

Neurons the specialized all of nervous system. In nervous system has no power of regeneration.

147)

Cardiac muscle fibers, also known as cardiomyocytes, are characterized by their striated, branched, and involuntary nature.

148)

NEW NCERT XI, Pg#110

149)

NEW NCERT XI, Pg#111

150)

NEW NCERT XI, Pg#111

151)

NEW NCERT XI, Pg#111

152)

NEW NCERT XI, Pg#111

153)

NEW NCERT XI, Pg#113

154)

NEW NCERT XI, Pg#112

155)

NEW NCERT XI, Pg#112

156)

NEW NCERT XI, Pg#113

157)

NEW NCERT XI, Pg#117

158)

NCERT XI,Pg#111

159)

NCERT XI,Pg#109

160)

NEW NCERT XI, Pg#117

161)

NEW NCERT XI, Pg#112

162)

Option 2 is correct

163)

NEW NCERT XI, Pg#107

164)

NEW NCERT XI, Pg#110NEW NCERT XI, Pg#107

165)

NEW NCERT XI, Pg#117

166)

NEW NCERT XI, Pg#117

167)

NEW NCERT XI, Pg#106

168)

NEW NCERT XI, Pg#112

169) Module-3 Pg # 127

170) NCERT-XI Eng Pg. # 157 (Last paragraph)

171) Module-3 Pg. # 130

172) Module-3 Pg. # 130

173)

NEW NCERT XI, Pg#117

174)

NEW NCERT XI, Pg#117

175)

NEW NCERT XI, Pg#117

176) NCERT XI Pg.# 146

177)

NEW NCERT XI, Pg#105

178)

NEW NCERT XI, Pg#105

179)

NEW NCERT XI, Pg#110

180)

NEW NCERT XI, Pg#110